

Classification models

Module 3 - Lesson 2

Overview

Classification assigns inputs to categories. Logistic regression, decision trees, and random forests are the standard tools, evaluated with accuracy, precision, and recall.

Key Concepts

- Logistic regression outputs class probabilities via the sigmoid function.
- Decision trees split on features to separate classes; random forests average many trees.
- Accuracy hides problems on imbalanced data; use precision, recall, and F1.
- The confusion matrix shows exactly which classes get confused.
- ROC/PR curves let you trade precision against recall via a threshold.
- Class imbalance needs resampling or class weights, not just accuracy.

Example

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import classification_report, confusion_matrix

model = RandomForestClassifier(n_estimators=200, random_state=42)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print(classification_report(y_test, y_pred))
print(confusion_matrix(y_test, y_pred))
```

Summary

Classification predicts categories. Choose the right model, evaluate with precision/recall on imbalanced data, and read the confusion matrix to understand real mistakes.

Practice Checklist

- Train a random forest and report precision and recall per class.
- Explain when accuracy is misleading and what to use instead.
- Plot the confusion matrix and identify the most common error.